BabyAI: A Platform to Study the Sample Efficiency of Grounded Language Learning

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github.com/mila-iqia/babyai

openreview.net/forum?id=rJeXCo0cYX

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Key points / Takeaways

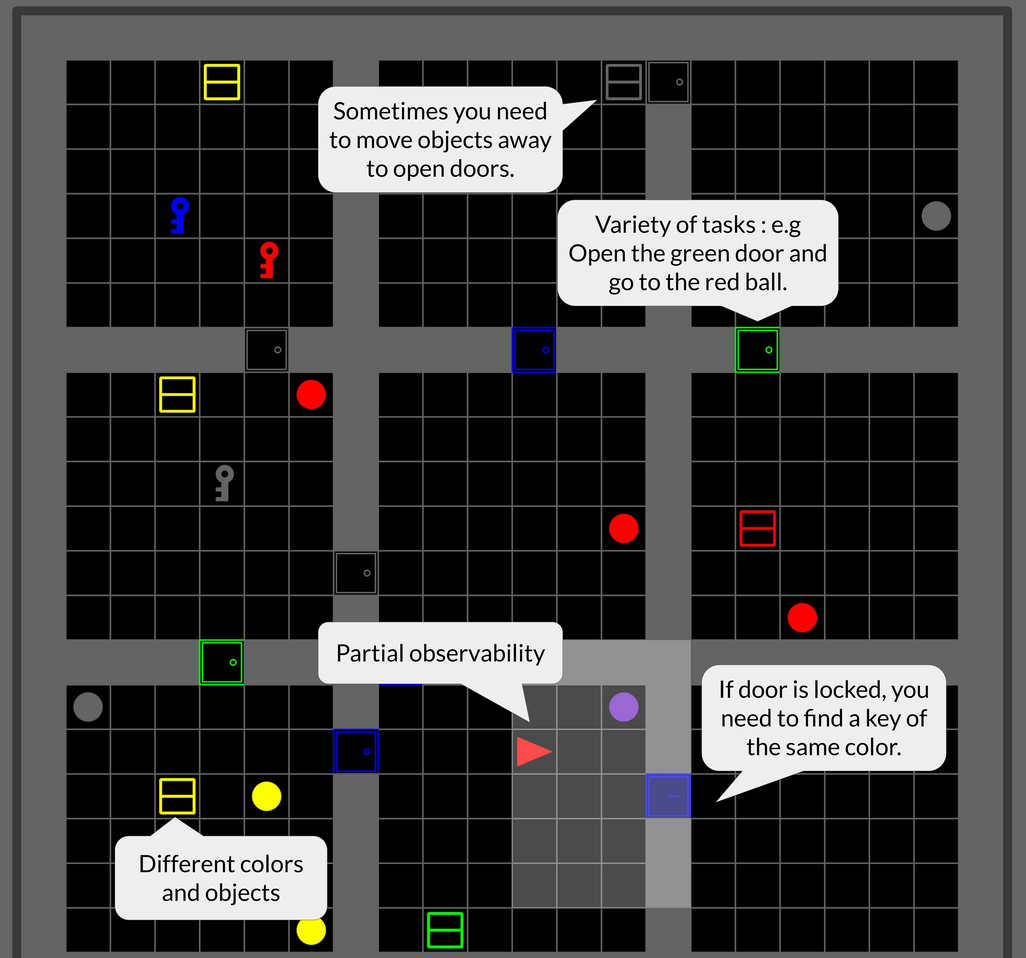
Benchmark for grounded language learning

- The CLEVR for instruction following
- Compositional Baby language

Simple but challenging tasks

- Gradual progression to learn Baby Language
- 19 levels of increasing difficulty
- Partial observability

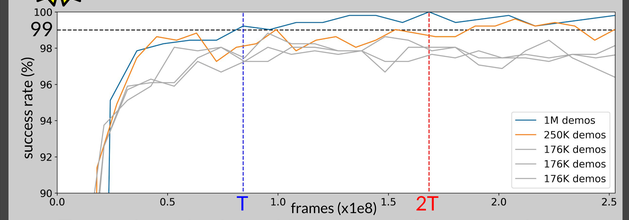
Fast & lightweight simulation
Compatible with OpenAI gym
Easily customizable environments



Baby language instructions

- `go to the red ball`
- `open the door on your left`
- `put a ball next to the blue door`
- `open the green door and go to the key behind you`
- `put a ball next to a grey box and pick up the yellow box`

Success criterion



T set when the job with the most demonstrations crosses the 99% threshold

Level solved if success rate > 99% before 2T frames

Baselines

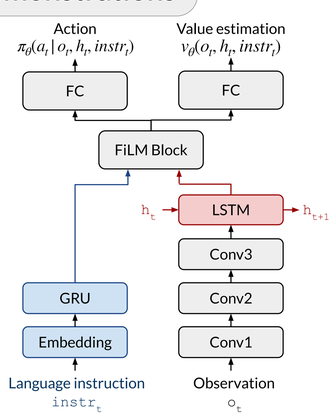
BabyAI Bot : heuristic agent that can solve every level

RL : Agent trained with PPO

Imitation Learning : Behavioral Cloning from Bot or RL demonstrations

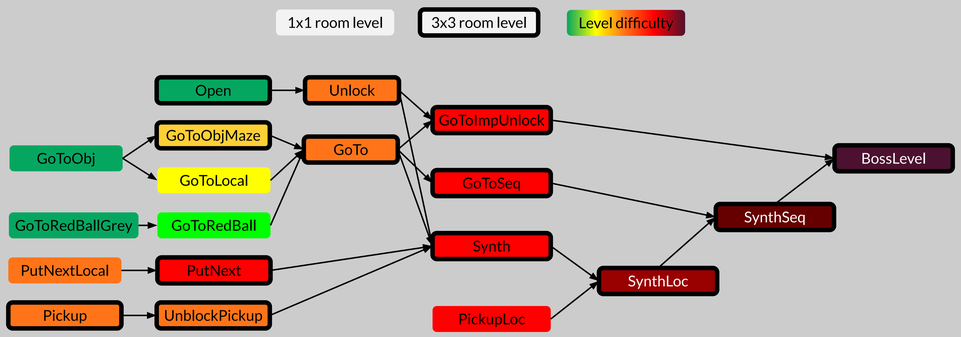
Level	IL from Bot	RL
GoToRedBallGrey	5.7 - 8	377 - 379
GoToRedBall	44.- 62	453 - 470
GoToLocal	125 - 177	1167 - 1320
PickupLoc	250 - 354	2591 - 2608
PutNextLocal	354 - 500	1875 - 2587
GoTo	250 - 354	1057 - 2177

Data efficiency of IL and RL as the number of episodes (in thousands) required to solve each level



Neural net architecture of the policy

The 19 levels of BabyAI



Curriculum learning

Pretrain on base level

Additional training on the target level

Base Levels	Target Level	With Pretraining	Without pretraining
GoToLocal	GoTo	250 - 354	250 - 354
GoToObjMaze	GoTo	354 - 500	250 - 354
GoToLocal and GoToObjMaze	GoTo	88 - 125	250 - 354
GoToLocal	PickupLoc	177 - 250	250 - 354
GoToLocal	PutNextLocal	250 - 354	354 - 500

Data efficiency results (in thousands) for pretraining experiments